TechMart Semantic Search: Build and Validate

Model: TF-IDF + SVD (offline stand-in, NOT all-MiniLM-L6-v2) | FAISS IndexFlatL2 (384-d) | 5000 products | 50 test queries | 2026-10-04

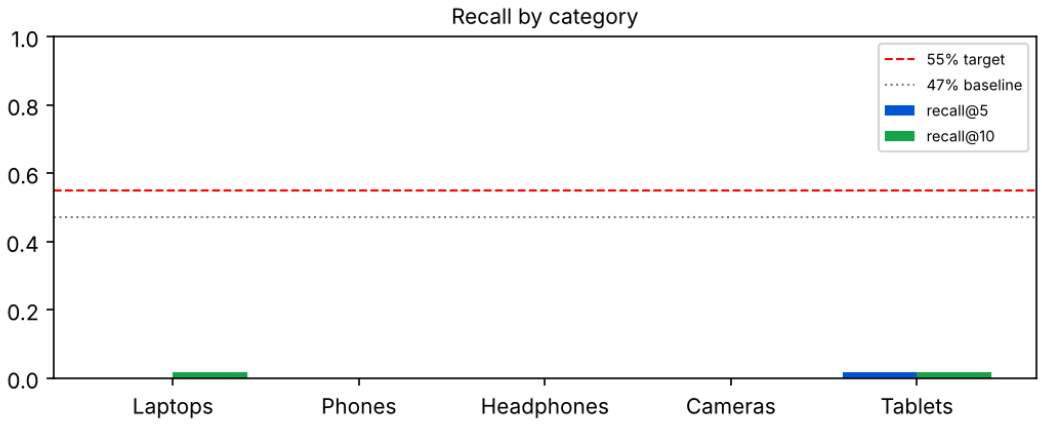
SAMPLE - generated with an offline TF-IDF stand-in because all-MiniLM-L6-v2 could not be downloaded in the build environment. These are NOT the results the lab requires. Re-run python techmart_search.py with internet access and submit that report.

1. Recall Metrics Report

Overall **recall@5 = 0.3%** and **recall@10 = 0.7%** across 50 queries. Against the 47% keyword baseline recall@5 is -46.7 points; it **does not meet** the 55% target (-54.7 points). Note: 84% of queries have more than 5 relevant items, so recall@5 cannot reach 100% for them.

Interpretation caveat. Each query's ground truth lists on average only 6.6 products, about 0.7% of the roughly 1000 catalog items in that category, and catalog descriptions are near-identical templates. Many unlabelled products are equally good matches, so recall can be low even when results are sensible. As a complementary check, **category precision@5** (top-5 results in the intended category) is **93.2%**.

Category	Queries	recall@5	recall@10	vs 55% target
Laptops	10	0.0%	1.7%	below
Phones	10	0.0%	0.0%	below
Headphones	10	0.0%	0.0%	below
Cameras	10	0.0%	0.0%	below
Tablets	10	1.7%	1.7%	below
Overall	50	0.3%	0.7%	below



2. Failure Analysis

50 of 50 queries scored recall@5 below 40%. Failures by category: Laptops 10, Phones 10, Headphones 10, Cameras 10, Tablets 10.

Query	Category	recall@5	wrong-cat in top5	# relevant
laptop for gaming	Laptops	0%	0	5
portable notebook computer	Laptops	0%	0	8

Query	Category	recall@5	wrong-cat in top5	# relevant
lightweight laptop	Laptops	0%	0	6
business laptop	Laptops	0%	0	6
student laptop	Laptops	0%	0	7
ultrabook thin	Laptops	0%	0	7
15 inch laptop	Laptops	0%	0	6
laptop with long battery	Laptops	0%	0	8

Pattern 1: Cross-category confusion / lexical overlap. 7% of top-5 results for failed queries came from the wrong category. Root cause candidate: generic catalog text is near-identical across categories, so the embedding keys on shared phrases. Fix: filter or boost by predicted category, or add category to the text.

Pattern 2: Poor or minimal descriptions. 8% of missed relevant products have minimal descriptions versus 9% of the catalog. A share well above the base rate points to data quality. Fix: enrich descriptions or fall back to title and attribute fields.

Pattern 3: Query wording: synonyms and length. Failed queries average 2.6 words versus nan for passing ones. Synonym or multi-attribute phrasing (for example 'notebook' for laptop) is a model limitation of a small general-purpose encoder. Fix: query expansion, a domain fine-tuned model, or hybrid BM25 + vector search.

Causes are inferred from the statistics above; confirm by reading the specific failures in outputs/query_results.csv.

3. Production Readiness Assessment

Recall@5 of 0.3% does not meet the 55% target, so the prototype is not ready for production and needs the fixes above first. Because the labelled set covers a small slice of equally relevant items, the recall target should be re-validated with fuller relevance labels before it gates a launch. Insights: recall@k must be read with the number of relevant items per query; category-level breakdowns reveal weaknesses that the average hides. Limitations: only 50 queries, ground truth of unknown labelling quality, an exact (flat) index that will need approximate search at larger scale, and no latency or freshness testing. Next steps: add hybrid lexical search, category-aware reranking, query expansion, a larger labelled set, an online A/B test against keyword search, and monitoring of recall and zero-result rate.